

AARON FOOTE

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EDUCATION

Wesleyan University, Middletown, CT

September 2024 - May 2025

M.A. in Computer Science

GPA: 4.22/4.00

Thesis: *Diagnosing Biased Feature Importance Scores in Tree-Based Models*

Wesleyan University, Middletown, CT

September 2020 - May 2024

B.A. with High Honors in Computer Science, Minor in Data Analysis

GPA: 3.87/4.00

Thesis: *On the Complexity and Threshold Behavior of Playing Nonogram Puzzles*

Relevant Coursework:

Mathematical Statistics, Statistical Learning, Algorithms, Survival Analysis, Applied Bayesian Analysis, Artificial Intelligence, Applications of Linear Algebra in Machine Learning, Software Engineering, Probability Theory, Data Structures

WORK EXPERIENCE

Founding Engineer

June 2025 - Present

Climecast

Palo Alto, CA

- Construct database for early-stage real estate startup, helping businesses meet regulations and reach their climate goals
- Develop user-friendly emissions and energy dashboards for properties using Python, SQL, and Typescript
- Build climate risk evaluation tools to protect home buyers

Data Engineering Consultant

June 2025 - Present

CMC Foundation for Change

Palo Alto, CA

- Implement data infrastructure to streamline processing of survey data with Python, SPSS,
- Develop data visualizations, reports, and dashboards for substance abuse therapy nonprofit

RESEARCH EXPERIENCE

Graduate Research Assistant – Interpretable ML

June 2024 - Present

Wesleyan University

Middletown, CT

- Developed a hypothesis test under minimal assumptions to evaluate feature importance methods for bias
- Validated the method through extensive simulation and application to real datasets in Python and R
- Integrated dimension reduction techniques to extend the test to higher dimension
- Investigated applications to gene expression data to produce reliable, efficient feature importance scores to improve the interpretability of black-box ML models for their application in medicine

Research Assistant – Algorithms, Computational Complexity

September 2022 - May 2025

Wesleyan University

Middletown, CT

- Established complexity results and phase transition behavior for the solving of combinatorial puzzles
- Developed solvers in C/bash for high-performance parallel computing cluster
- Studied the complexity of puzzles published online, scraping thousands of published puzzles with Python
- Modeling human solving strategies to understand the connection between human and computational difficulty
- Formalizing and proving complexity results for problems related to generating puzzles

Summer Research Fellow – Computer Vision, Natural Language Processing

June 2021 - September 2021

Wesleyan University

Middletown, CT

- Developed a pipeline to extract key frames from political ads using corner detection algorithms and OCR, cutting the financial cost of analyzing political ad data by a factor of 100
- Implemented an algorithm to clean output from OCR into time-ordered, repeat-free output using string clustering
- Attended statistical methods workshops covering log-linear modeling (Poisson, negative-binomial, ordinal), longitudinal analysis, regression post-estimation, survival analysis, and exploratory data analysis

TEACHING EXPERIENCE

Head Quantitative Analysis Tutor

Wesleyan University

April 2023 - May 2025

Middletown, CT

- Hold 10+ hours of weekly office hours open to the entire university, assisting with coursework, projects and research
- Provide one-on-one statistical consulting for undergraduate and graduate thesis projects, ensuring methodological rigor
- Guide students in managing and analyzing data, employing and interpreting statistical techniques, and visualizing their work with R, STATA, Python, SPSS, SAS, and Excel
- Topics tutored: data visualization, survival analysis, machine learning, hypothesis testing, regression, regression diagnostics, longitudinal data analysis, econometrics, data engineering
- Develop a schedule each semester, lead weekly meetings, and consult with professors

Statistics/Data Analysis Teaching Assistant

Wesleyan University

January 2024 - May 2024

Middletown, CT

- **Survival Analysis** (Spring 2024): Clarified concepts in class, guided students on lab work, and consulted with students on final projects involving topics such as censoring, hazard and survival functions, Kaplan Meier estimates, Cox proportional hazards model, time-varying covariates, parametric survival analysis, and accelerated failure-time distributions
- **Applied Data Analysis** (Spring 2024): Mentored six students for five hours per week at every stage of a research project (identifying research questions, preparing and visualizing data, running statistical tests, and linear/logistic regression)

Computer Science Teaching Assistant

Wesleyan University

September 2022 - December 2024

Middletown, CT

- **Algorithms** (Fall 2024): Instructed a dozen students once a week on material including analysis of algorithms, divide-and-conquer, dynamic programming, greedy algorithms, branch and bound, and approximation algorithms
- **Theory of Computation** (Fall 2022): Instructed students once per week on material including finite automata, Turing machines, NP-completeness, polynomial time reductions, space complexity, and approximation algorithms

RESEARCH PRESENTATIONS

Foote, A. and Krizanc, D. (2025) "Nonogram: Complexity of Inference and Phase Transition Behavior." *arXiv preprint* doi:10.48550/arXiv.2507.07283

Foote, A. and Krizanc, D. (2025) "TRIP: A Nonparametric Test to Diagnose Biased Feature Importance Scores." *IJCAI Workshop on Explainable Artificial Intelligence* (August 2025, Oral Presentation) doi:10.48550/arXiv.2507.07276

Foote, A. and Matthews, G. (2024). "Continuous Within-Play Estimation of Run-Stopping Contribution in American Football." *Cascadia Symposium on Statistics in Sports* (September 2024, Conference Poster).

Foote, A., Gooyabadi, M. and Addleman, N. (2023). "Factors in Learning Dynamics Influencing Relative Strengths of Strategies in Poker Simulation." *Games*. doi:10.3390/g14060073.

AWARDS

Maureen Snow Prize (July 2023): Runner-up among 200 students, given for most effective communication of research to a broad audience

Baker '64 Collabria Fellowship in Data Analysis (April 2023) – \$5,000

Best Data Story Award (April 2023): Awarded at Northeast Division ASA DataFest Competition

QAC Apprentice Grant (April 2021) – \$5,400